Energy Efficiency 2025

Buildings

How and where is energy used?

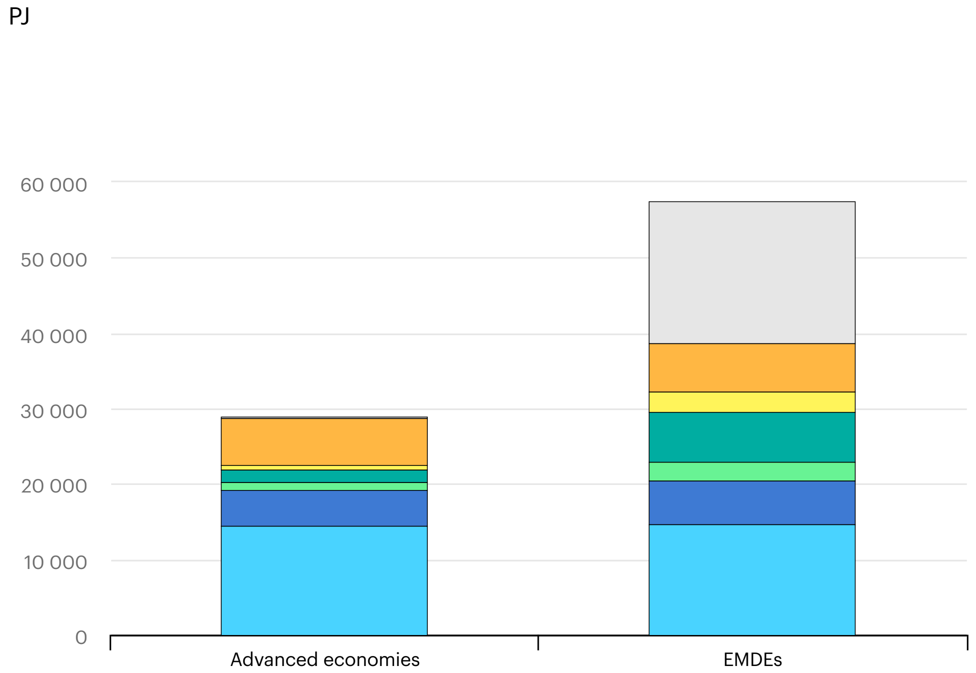
Total final consumption in 2024 was over 450 EJ and has grown by around 25 EJ since 2019. Buildings account for around 30% of global energy demand and have contributed around 20% of the growth in total demand since 2019. The residential sector makes up about 70% of total energy demand in buildings, while the remaining 30% is used in commercial and public buildings.

In advanced economies, most energy in homes is used for space and water heating, together accounting for about 70%. This is followed by the use of electrical appliances, such as refrigerators, televisions and washing machines. Addressing inefficient heating technologies and poorly insulated buildings is key in most advanced economies to accelerate efficiency progress. Space cooling is still a minor share of total demand, but it is expected to grow in the coming years *(https://www.iea.org/commentaries/staying-cool-without-overheating-the-energy-system)*.

In emerging and developing economies, per capita space and water heating energy use is less than one-fourth that of advanced economies, due to a large share of the population living in warmer climates and lower access to affordable energy services. Growing populations, rising incomes and hotter summers *(https://www.iea.org/commentaries/data-tool-gives-fine-grained-view-of-climate-vulnerabilities-in-the-energy-system-and-beyond)* are increasing the total residential energy demand in EMDEs, however, particularly for space cooling. Addressing inefficient cooling technologies, ensuring new buildings are efficiently designed and constructed, and promoting clean cooking can help emerging and developing economies to accelerate efficiency progress.

Total energy demand in the residential sector, by end-use, advanced economies and emerging economies, 2023

Open ↗



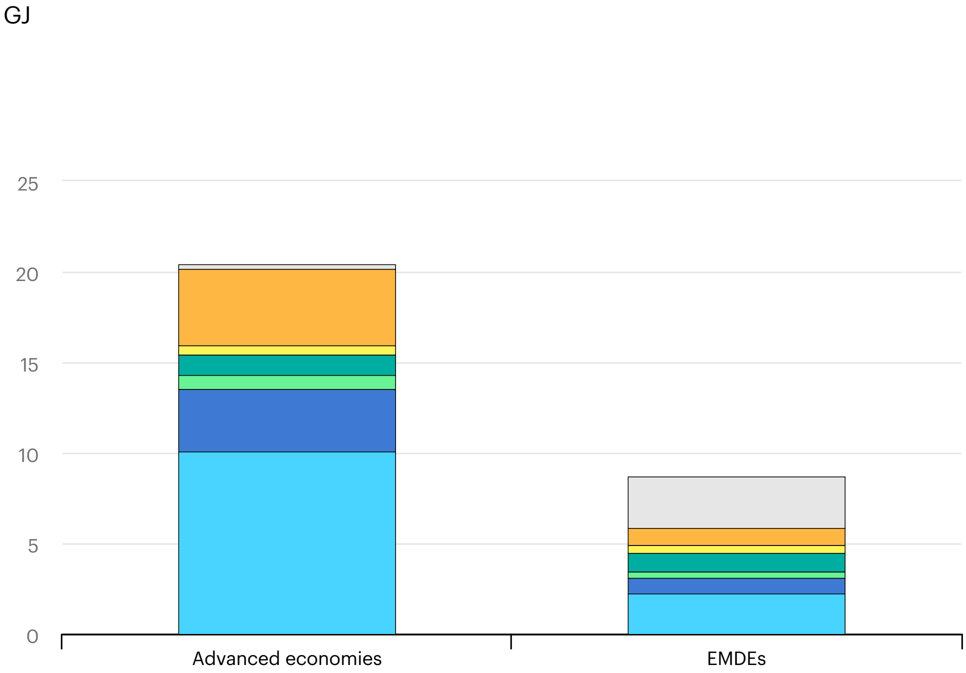
IEA. Licence: CC BY 4.0 *(https://www.iea.org/terms/creative-commons-cc-licenses)*

Space heating · Water heating · Space cooling · Cooking · Lighting · Appliances · Traditional use of biomass

Notes

Per capita energy demand in the residential sector, by end use, in advanced and emerging economies, 2023

Open ↗



IEA. Licence: CC BY 4.0 (https://www.iea.org/terms/creative-commons-cc-licenses)

Space heating · Water heating · Space cooling · Cooking · Lighting · Appliances · Traditional use of biomass

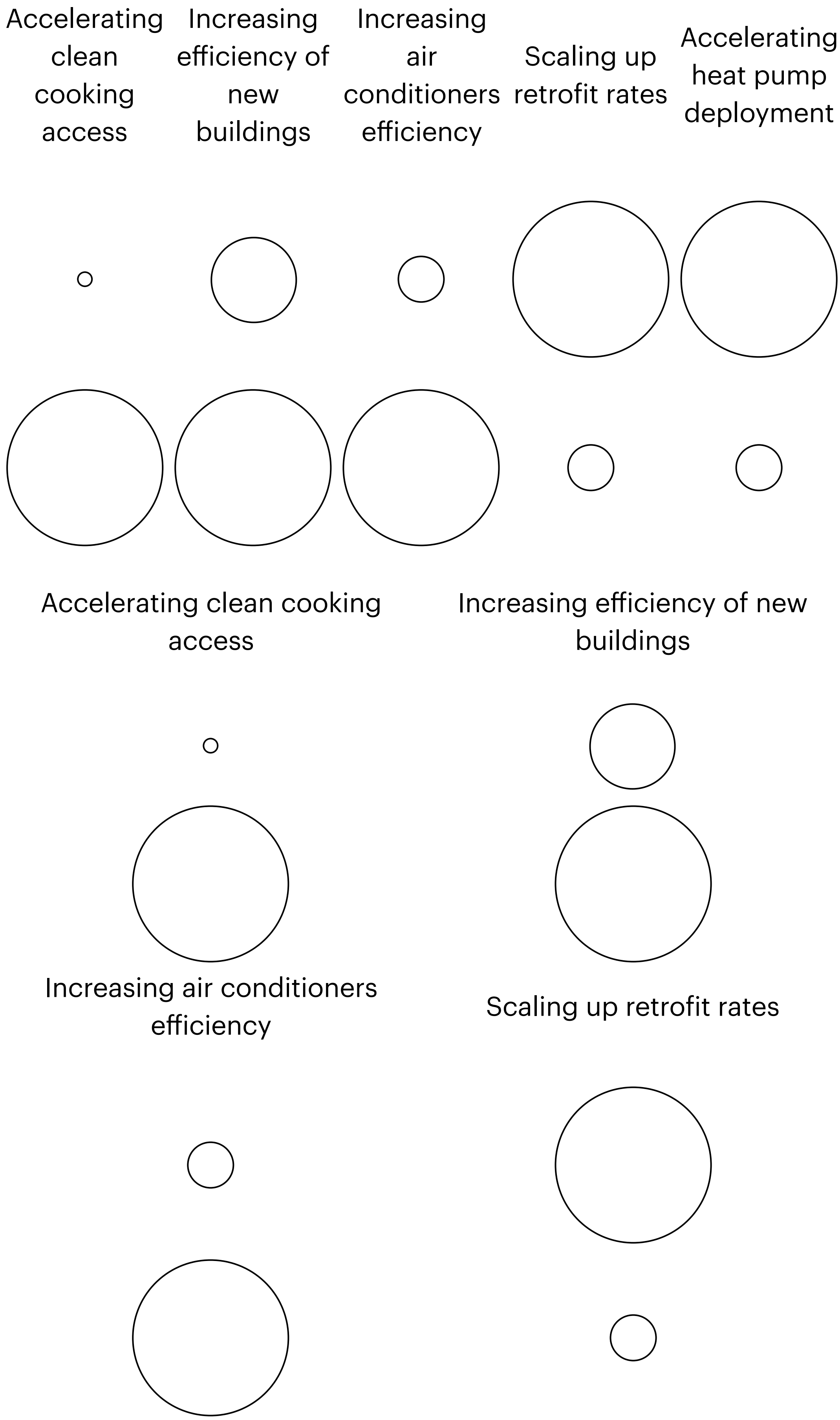
Notes

What are the key measures to accelerate progress?

In advanced economies, the number of buildings is growing at a slow rate, and many existing buildings are poorly insulated. Accelerating retrofits to reduce heating and cooling energy demand, and electrifying heating systems, are therefore some of the most important drivers of efficiency progress. Policies, such as energy performance certificates *(https://www.iea.org/reports/energy-efficiency-policy-toolkit-2025/energy-efficiency-policy-toolkit-case-studies)* and retrofit incentives can help improve the efficiency of existing buildings. To accelerate the deployment of electric heating systems, such as heat pumps, governments can regulate fossil fuel-based heating technologies, harmonise labels for heating solutions, provide targeted rebates, or reduce the price gap *(https://www.heatpumps.org.uk/new-european-heat-pump-market-report-highlights-need-to-solve-uks-electricity-and-gas-price-disparity/)* between gas and electricity.

In emerging economies, the number of buildings is growing almost twice as fast as in advanced economies, so efficient new buildings and urban planning are important to accelerate efficiency progress. Building energy codes, which currently apply to around half of new floor area in emerging economies *(https://www.iea.org/reports/energy-efficiency-2024)*, can ensure that new buildings are built efficiently. Improving the efficiency of air conditioners is another key focus area, given the high and increasing *(https://www.iea.org/commentaries/data-tool-gives-fine-grained-view-of-climate-vulnerabilities-in-the-energy-system-and-beyond)* need for cooling, and the strong growth in sales of air conditioners *(https://www.jraia.or.jp/english/statistics/)*. Lastly, reducing the use of traditional biomass through clean cooking measures can prevent millions of premature deaths *(https://www.iea.org/reports/universal-access-to-clean-cooking-in-africa)*, while increasing in cooking efficiency.

Key measures to accelerate efficiency progress in the residential buildings sector



Accelerating heat pump deployment

Advanced economies
Emerging markets and developing economies

Advanced economies
Emerging markets and developing economies

Notes: Bubble sizes give an indication of the relevance of each measure within the country grouping (classified as high, moderate, low and very low). The classification is based on multiple factors, such as how and where energy is used, current trends and current policy coverage. Actions listed are not exhaustive but are meant to give policy makers an overview of key focus areas that can accelerate progress.

Next
Transport *(/reports/energy-efficiency-2025/transport#abstract)*